Bioinformatics core facility management training – availability and challenges

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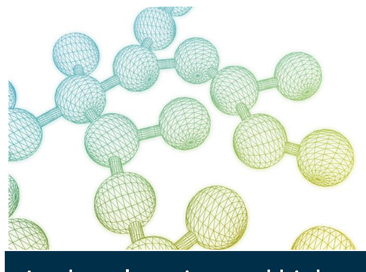
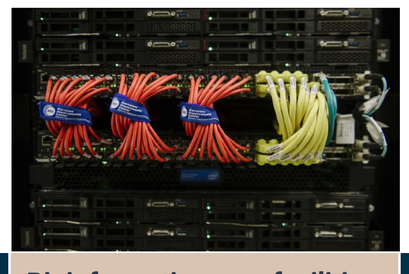
Trainings supported by

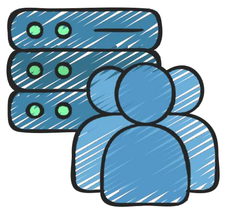
Bioinformatics Core Facilities



As deep learning and high throughput sequencing methodologies are increasingly applied to analyse molecular data, there is a growing need for highly specialised services from Bioinformatics Core Facilities and their supporting teams.

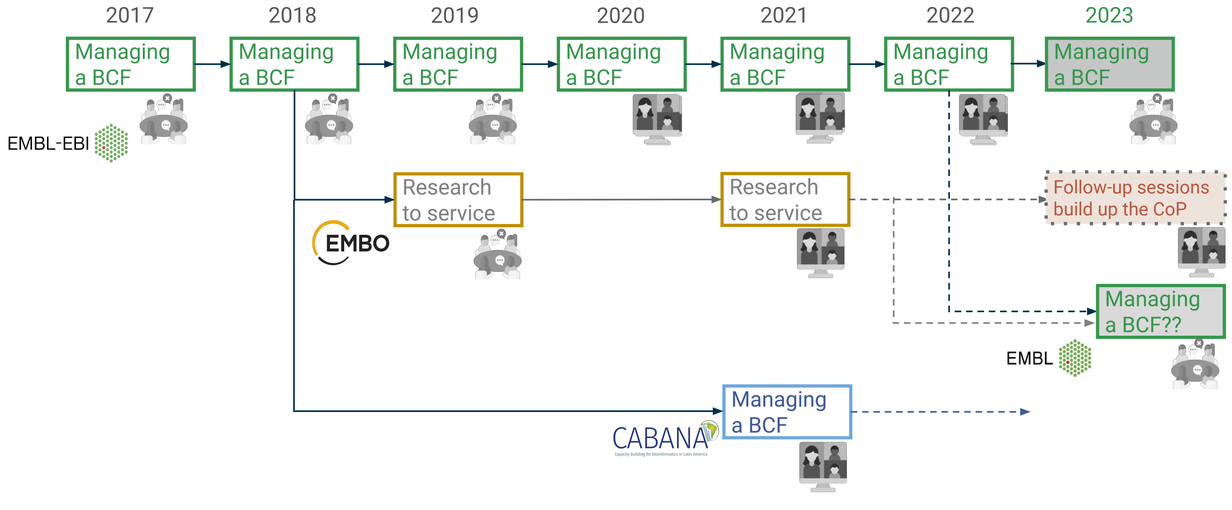


Bioinformatics core facilities play an essential role in enabling research in life sciences



Added to these challenges, the rapidly changing necessities of biological data analysis quickly outgrow the capacities of these facilities in terms such as software or hardware usage.

Timeline



Bioinformatics Core Facilities management courses content

LEARNING OUTCOMES	Managing a Bioinformatics Core Facility (EMBL-EBI)	From research to service: planning and running a Bioinformatics Core Facility (EMBO)
Identify the **strengths and limitations** of the Bioinformatics Core Facility	✓	✓
Apply a framework for **designing new services** or redesigning existing services (including training users)	✓	✓
Estimation of **computing and human resources**. Appreciate different **cost models**, their pros and cons, and how to apply them in your context	✓	✓
Data management: user support - submission, and sensitive data	✓	✓
COURSE LENGTH	Three-day course with follow-up sessions according to groups needs	Five-day course with a follow-up session
AUDIENCE	Senior staff members or managers	Individuals transitioning from research roles to Bioinformatics Core Facility management positions, or more junior scientists

Key points from past iterations

The courses in numbers
- >100 participants
- >200 applicants
- 28 trainers (Research to service – near 1:1 trainer/trainee ratio)

The ideal modality is face-to-face, incentivises networking within:
- Participants
- Trainers (participants become trainers)
- Principal investigators in Life Sciences

Content update iteration and adaptation on the fly according to participant's challenges and needs.

Frequently Bioinformatics Core Facilities are small, in special the ones starting. The networking and sharing experiences aspects are crucial:
- Sessions with ISCB Bioinfo-Core
- Regular short events with diverse topics of discussion
- Foster a Community of Practice across the training iterations

Sustainability requires multiple soft skills… negotiation!
- Technical and soft skills, particularly management competencies

Career progression, which in some extent may be due to little awareness of competency frameworks

More in-depth training opportunities for BCF scientists

1) **EMBL visitor programme** - there are opportunities for secondments/sabbaticals to an EMBL group (including EBI services and EMBL core facilities) - contact the group leader you'd like to visit directly. https://www.embl.org/about/info/scientific-visitor-programme/fellowships
2) **The ARISE programme**. This takes training for research infrastructure a step further - you can spend 3 years in a core facility or another EMBL service developing your competency as a research infrastructure specialist. Not limited to bioinformatics but there are plenty of bioinformatics services participating.
3) **EMBO** is now offering **core facility fellowships** https://www.embo.org/funding/fellowships-grants-and-career-support/core-facility-fellowships
4) **RItrainPLus staff visits and secondments** - https://ritrainplus.eu/staff-knowledge-exchange-programme/
5) **Managing a Bioinformatics Core Facility** online training - 23-25 Nov (EMBL-EBI) - Apply before 28 Sept https://www.ebi.ac.uk/training/events/managing-bioinformatics-core-facility-2022

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